PRACTICAL 2. Core JavaScript Concept for MERN

PRACTICAL NO.2(A)

Aim:- Create a function and call them with parameter.

Code:-

```
//Function with Parameters
function greet(name){
   console.log("Hello,"+name);
}

//Function Call
greet("John");
greet("Alice")
```

Output:-

```
Hello,John
Hello,Alice
```

PRACTICAL NO.2(B)

Aim:- Work with arrays and object in JavaScript.

Code:-

```
//Array
let fruits=["Apple","Banana","Mango"];

//Object
let students={
   name:"John",
   age:20,
   course:"TYBSc IT"
};
```

```
console.log("Fruits:",fruits);
console.log("Student Name:",students.name);
console.log("Age:",students.age);
console.log("Course:",students.course);
```

Output:-

```
Fruits: [ 'Apple', 'Banana', 'Mango' ]
Student Name: John
Age: 20
Course: TYBSc IT
```

Push & Pop Array Method:-

Code:-

```
let students=["Amit","Sneha","Rohan"];

//Array Methods
students.push("Priya");
students.pop();

console.log("Students List:");
console.log(students);

let Student={
   rollNo: 101,
   name: "Amit",
   course: "TYIT"
};

console.log("\nStudents Details");
console.log("Student RollNo:",Student.rollNo);
console.log("Student Name:",Student.name);
console.log("Student Course:",Student.course);
```

Output:-

```
Students List:
[ 'Amit', 'Sneha', 'Rohan' ]
```

Students Details:-
Student RollNo: 101
Student Name: Amit
Student Course: TYIT

PRACTICAL NO.2(C)

Aim:- Demonstrate use of arrow function.

Code:-

```
//Arrow Function
const add=(a,b)=>a+b;

console.log("Sum=",add(10,20));
```

Output:-

```
Sum= 30
```

PRACTICAL NO.2(D)

Aim:- Write a program using array methods (map, filter, reduce).

Code:-

```
let numbers=[10,20,30,40,50];

let doubled=numbers.map(num=>num*2)
let greaterThan25=numbers.filter(num=>num>25);
let total=numbers.reduce((sum,num)=>sum+num,0);

console.log("Original:",numbers);
console.log("Map:",doubled);
console.log("Filter:",greaterThan25);
```

```
console.log("Reduce:",total);
```

Output:-

```
Original: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
```

PRACTICAL NO.2(E)

Aim:- Create a program that manipulates objects and displays output.

Code:-

```
let Student={
   name:"Amit",
   age:20,
   course:"BCA",
   marks:78
};

//Add a new property
Student.grade="A";

//Update marks
Student.marks=85;

//Disply students details
console.log(Student);
```

Output:-

```
{ name: 'Amit', age: 20, course: 'BCA', marks: 85, grade: 'A' }
```

PRACTICAL NO.2(F)

Aim:- Implement a small JavaScript program combining function, arrays, and objects.

Code:-

```
//Array of objects
let students=[
    {name:"John",marks:85},
    {name:"Alice",marks:92},
    {name:"Rahul",marks:78}
];

//Function
function displayStudents(studentsList){
    studentsList.forEach(student=>{
        console.log(student.name+"-"+student.marks);
    });
}

//Function Call
displayStudents(students);
```

Output:-

```
John-85
Alice-92
Rahul-78
```